



VAMP 1.0 Production Readiness

Intelligent Architecture & Capability Review

North-West University • Performance Management Ecosystem • 2025 Academic Calendar Aligned

Avatar State Diagnostic

State 1: Idle (Ready State)



System Triggers: Page load, action completion, no activity for 3+ seconds.

UI Feedback: Awaiting instruction...

State 2: Busy (Cognitive Load)



System Triggers: File upload starts, AI queries submitted, report generation.

UI Feedback: Analysing... / Consulting the archives...

State 3: Speaking (Response Loop)



System Triggers: AI responses received, scan completes, error/success states.

Behavior: Auto-returns to Idle based on reading time (30ms per character).

Expectations Anatomy



Raw Task Agreement (TA) Import

KPA1: Teaching
(1337.62h / 79.14%)

KPA2: OHS

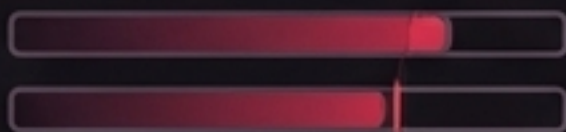
KPA3: Research

KPA4: Leadership
(203.0h / 12.06%)

KPA5: Social

Monthly Buckets

**Minimum Target vs.
Stretch Target**



Milestone Tasks

- Proposal Submission
- Progress Report 1
- Final Presentation

Addendum B

Teaching Modules
(HISE 411, HISE 322)

1 Contract expands into 100+ perfectly organized, trackable tasks.

The NWU 2025 Calendar Overlay

January

KPA1 (Semester 1 Prep, eFundi setup) + KPA3 (Ethics applications) + KPA4 (Annual planning).

March-May

KPA1 (Semester 1 Teaching peak) + KPA3 (NRF rating window in Mar).

June

KPA1 (Milestone: Semester 1 marks submission) + KPA3 (Mid-year output submission).

December

KPA1 (Year-end moderation).

JAN

FEB

MAR

APR

MAY

JUN

JUL

AUG

SEP

OCT

NOV

DEC

January

KPA1 (Semester 1 Prep, eFundi setup) + KPA3 (Ethics applications) + KPA4 (Annual planning).

June

KPA1 (Milestone: Semester 1 marks submission) + KPA3 (Mid-year output submission).

November

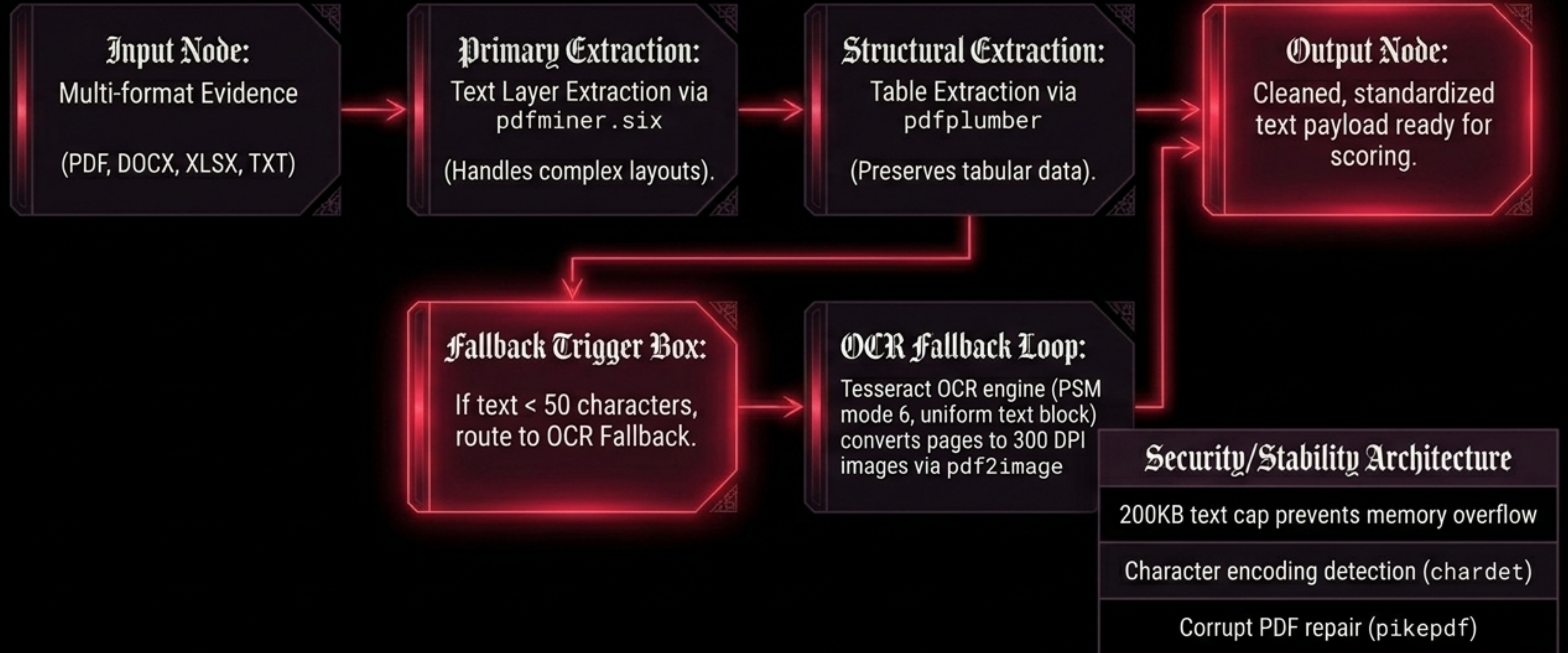
KPA3 (Milestone: Year-end accredited publication).

December

KPA1 (Year-end moderation).

VAMP does not just track tasks; it anticipates institutional rhythms, automatically scaling expectations to match active teaching, research, and governance cycles.

Multi-Stage Ingestion Pipeline



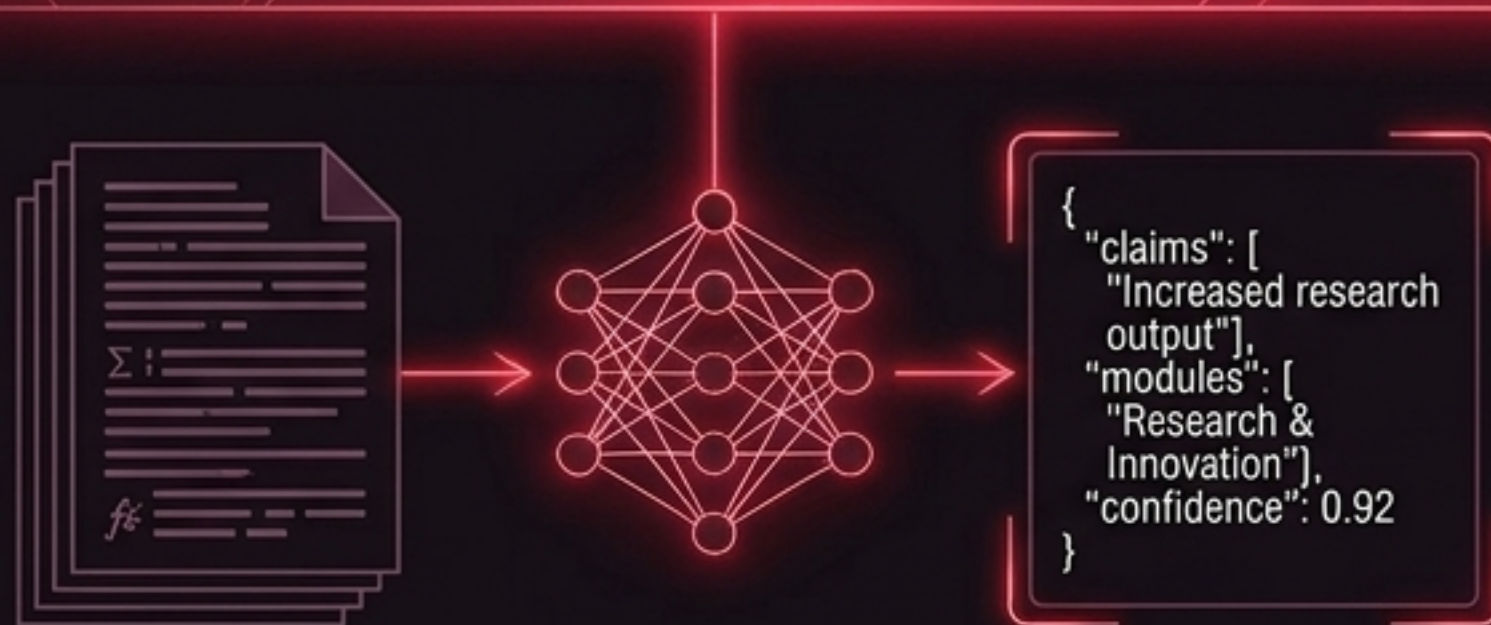
Dual-Pass Brain Scorer

Pass A: Semantic Extraction

Mechanism: LLM analyzes evidence to extract schema (claims, modules, confidence).

Constraint: Prompt strictly forbids assigning ratings or KPAs.

LLM proposes structure → Deterministic Engine enforces NWU taxonomy.



Pass B: Institutional Law

Mechanism: Hardcoded NWU rules process the extracted schema.

Actions:

- KPA Routing (5 KPAs + regex cues)
- Tier Classification (Transformational / Developmental / Compliance)
- Values Detection (15 core NWU values)
- Policy Recognition (133 institutional policies)



Math to Visual Cascade

Artefact Contribution Score (ACS)

$$\text{ACS} = (\text{Completion Estimate}) \times (\text{Credibility Weight, e.g., Rubric}=0.9) \times (\text{Confidence})$$

Capped at 1.0

KPI Completion Score (KCS)

$$\text{KCS} = \text{Sum of all ACS mapped to a specific KPI}$$

Status thresholds: ≥ 0.8 Achieved, ≥ 0.5 Partially Achieved

KPA Completion Rating (KCR)

$$\text{KCR} = \text{Average of KPI scores within a KPA}$$

Evidence sufficiency checks apply

Final Rating

$$\text{Final Rating} = \text{Weighted sum of all KCRs using specific Task Agreement weights.}$$

Confidence & Assertion Comparison Matrix

Algorithmic Guessing (Legacy)

Mechanism: Relies on keyword overlap.

Confidence: Sits at 35-50%.

Result: OHS_Certificate.pdf yields weak match.
Brain scorer reclassifies and overrides user intent.
Evidence gets lost.

VS

User-Driven Semantic Enhancement (VAMP 1.0)

Mechanism: User asserts mapping ("This proves fire safety training"). Groq/Ollama contextualizes the human description.

Logic update: Generic mapping is skipped.
Brain scorer override is skipped.

Confidence: Bumps to 85-95%.

Result: Evidence is stored with the exact user-selected KPA. The system completely respects the academic's assertion.

Outlook Integration Diagnostic



Passive Mailbox Scraping (Rejected)

Scraping whole mailboxes, storing raw dumps without relevance, running independent scoring outside VAMP.



Targeted Evidence Fetcher (Implemented)

Trigger: Driven by the current selected month and incomplete tasks.

Engine: Uses Playwright session state.

Search Plan: Derives queries from task titles, KPA codes, and evidence hints.

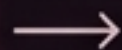
Ingestion: Pushes normalized attachments and email bodies through the exact same VAMP file ingest pipeline as manual uploads.

VAMP decides what is needed. Outlook simply fetches candidate evidence to fill specific, institutional gaps.

Contextual Awareness Features

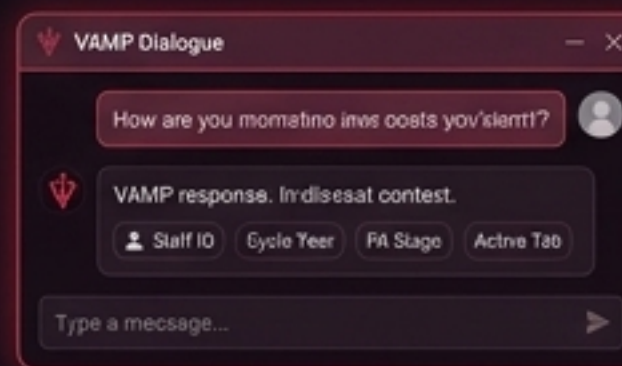
Feature 1: The Month-Closing Review

Check Month Status



Analyzes all evidence for the month, compares against minimum requirements, identifies gaps by KPA, and outputs actionable next steps (e.g., 'Upload 1 more module guide for HISE 411').

Feature 2: Context-Aware Chat (Ctrl+K)



VAMP intrinsically knows the user's Staff ID, the current Cycle Year, the active stage of the PA, and the exact tab they are viewing. Responses are never generic.

Absolute Validation

Automated Test Suite

31 / 31

Tests Passing



Covers Batch 7 Scorer (5),
Batch 8 Aggregator (6),
Batch 10 PA Generator (4),
Contract/Integration logic (16).

Security Assessment

0 / 0

Alerts Vulnerabilities

CodeQL Scan Result.
Hash-based secure file naming,
secure file paths, no user data
in system commands.

Offline Graceful Degradation



System degrades beautifully.
Deterministic scoring and core
library functions operate perfectly
offline and headlessly, even if LLM
or GUI resources are unavailable.



The Definitive Output: Perfect PA Generation

- Generates strictly to the NWU pa-report sheet layout.
- Preserves exact KPA ordering (1-6) and bold formatting.
- Inherits verified TA hours and weights.
- Multi-line Outputs, KPIs, and Outcomes rendered flawlessly.
- Validates total weight is $\approx 100\%$ before export.

VAMP 1.0 transcends standard evidence tracking. It is a secure, deterministically fair, and perfectly aligned performance architecture.